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Universal scaling laws and density slopes for dark matter haloes

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ABSTRACT

Small-scale challenges suggest some missing pieces in our current understanding of dark matter. In a bottom-up hierarchy, low-mass halos collapse early and retain strong primordial imprints, while prolonged assembly of large halos near m_h^* tends to erase that memory and establishes a universal spectrum. A transition range exists between the linear and nonlinear regimes, centered around m_h^*, within which gravitational dynamics dominate and self-organize the matter field into a universal spectrum with an effective index n=-1. A generalized "cascade", not restricted to viscous turbulence, can be used to describe the redistribution of mass and energy across scales, which enables the universality and universal scaling laws in this range [1]. In this paper, a cascade theory for dark matter is proposed to provide extra insights, similar to the cascade phenomenon in hydrodynamic turbulence. The kinetic energy is cascaded in dark matter from small to large scales involving a constant rate ε_u ($\approx -4.6\times10^{-7}m^2/s^3$). Confirmed by N-body simulations, the energy cascade leads to a two-thirds law for kinetic energy v_r^2 on scale r such that $v_r^2 \propto (\varepsilon_u r)^{2/3}$. Equivalently, a four-thirds law can be established for halo density ρ_s enclosed in the scale radius r_s such that $\rho_s \propto \varepsilon_u^{2/3} G^{-1} r_s^{-4/3}$, which can be confirmed by galaxy rotation curves. The critical properties of dark matter might be obtained by identifying key constants on relevant scales. First, the largest halo scale r_l can be determined by $-u_0^3/\varepsilon_u$, where u_0 is the dispersion of the velocity. Second, the smallest scale r_η depends on the nature of dark matter. For collisionless dark matter, the length scale $r_\eta \propto (-G\hbar/\varepsilon_u)^{1/3} \approx 10^{-13}m$ was found along with the mass scale $m_X \propto (-\varepsilon_u\hbar^5 G^{-4})^{1/9} \approx 10^{12}GeV$, where $\hbar$ is the Planck constant. An uncertainty principle for momentum and acceleration fluctuations is also postulated. For self-interacting dark matter, $r_\eta \propto \varepsilon_u^2 G^{-3}(\sigma/m)^3$, where σ/m is the cross section of interaction. On halo scales, the energy cascade leads to an asymptotic density slope $\gamma = -4/3$ for fully virialized haloes with vanishing radial flow, which might explain the nearly universal halo density. Based on the continuity equation, the halo density is analytically shown to be closely dependent on the radial flow and mass accretion, so that simulated haloes can have different limiting slopes. A modified Einasto density profile is proposed.

Key words: Dark matter flow; N-body simulations; Rotation curve; Collisionless; Self-interacting; Core-cusp; Energy cascade

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1 INTRODUCTION

Standard CDM (cold dark matter) paradigm of cosmology has many successes in the formation and evolution of large-scale structures and the contents and states of our universe [2, 3, 4, 5]. Despite great successes, serious theoretical and observational difficulties still exist [6, 7]. In particular, the predictions of the CDM model of structures on small scales (<1Mpc) are inconsistent with some observations. Examples are the core-cusp problem [8, 9], the missing satellite problem [10, 11], the too-big-to-fail problem [12, 13]. In addition, the origin of the baryonic Tully–Fisher relation (BTFR) and MOND (modified Newtonian dynamics) [14, 15, 16] is still not clear.

These small-scale challenges might be related to each other [7, 17, 18] and suggest missing pieces in our current understanding. First, the cusp-core problem describes the discrepancy between the cuspy halo density predicted by cosmological CDM-only N-body simulations and the cored density inferred from observational data for dwarf galaxies. The predicted halo density exhibits a cuspy profile with inner density $\rho(r) \propto r^\gamma$, where the slope γ persistently exceeds different observations [19, 20, 21, 22]. Even for the cuspy profile predicted by cosmological simulations, there seems no consensus on the exact value of the asymptotic slope γ, but with a wide range between -1.0 and -1.5. Since the first prediction of $\gamma = -1.0$ in the NFW profile [23], the inner density slope of simulated haloes appears to have different values from $\gamma > -1.0$ [24] to $\gamma = -1.2$ [25], and $\gamma = -1.3$ [26, 27]. To summarize, some key questions are: Is there an asymptotic slope for dark matter haloes? Why exists a nearly universal density profile? Why do different inner slopes γ exist in simulations?

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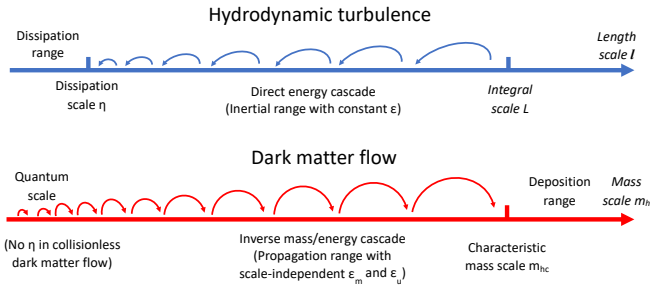


Figure 1. Schematic plot of the direct energy cascade in turbulence and the inverse mass and energy cascade in dark matter flow. Haloes merge with single mergers to facilitate a continuous mass and energy cascade to large scales. Scale-independent mass flux ε_m and energy flux ε_u are expected for haloes smaller than a characteristic mass scale (propagation range similar to the inertial range in turbulence). Mass cascaded from small scales is consumed to grow haloes at scales above the characteristic mass (the deposition range similar to the dissipation range in turbulence), where mass and energy flux become scale-dependent [1, 41, 42].

The halo density inferred from observational data exacerbates the problem. Even the smallest predicted inner-density slope from simulations is still greater than that from observations. Many solutions have been suggested to solve the cusp-core problem [28]. Within the CDM framework of collisionless dark matter, baryonic solutions focus on different mechanisms for energy exchange between baryons and dark matter to enable a flatter inner density [29, 30, 31]. Beyond the CDM framework, self-interacting dark matter is proposed as a potential solution [32, 33, 34]. The elastic scattering with a given cross section facilitates the exchange of momentum and energy between dark matter particles and the formation of a flat core. Although the existence of dark matter is supported by numerous astronomical observations [35, 36], the nature and fundamental properties of dark matter are still a big mystery. No matter whether collisionless or self-interacting, some key questions remain open: What are the limiting length or density scales for dark matter if they exist? What is the effect of self-interaction on these scales? What are the fundamental properties (particle mass, cross section, etc.) of dark matter? Answers to these questions would be critical for identifying and detecting dark matter.

In this paper, a cascade theory for dark matter flow is proposed to provide some useful insights, similar to the cascade in hydrodynamic turbulence. Both dark matter flow and turbulence are typical nonequilibrium systems involving energy cascade as a key mechanism to continuously release energy and maximize system entropy. To grasp the key idea, we first present the cascade in turbulence that has been well studied for many decades [37, 38, 39]. As shown in Fig. 1, turbulence consists of a collection of eddies (building blocks) on different length scales l that interact with each other. The classical picture of turbulence is an eddy-mediated energy cascade process, where the kinetic energy of large eddies feeds smaller eddies, which feeds even smaller eddies, and so on to the smallest scale η where viscous dissipation is dominant. The direct energy cascade in turbulence can be best described by a poem [40]:

"Big whirls have little whirls, That feed on their velocity;
And little whirls have lesser whirls, And so on to viscosity."

Despite the similarities, dark matter flow exhibits many different behaviors due to its collisionless and long-range interaction nature. First, unlike turbulence that is incompressible on all scales, dark matter flow exhibits scale-dependent flow behaviors, i.e. a constant divergence flow on small scales and irrotational flow on large scales [43]. Second, long-range gravity requires the formation of a broad spectrum of haloes to maximize the entropy of the system [44]. In principle, haloes of different masses can be grouped into groups of haloes with the same mass m_h. Mass accretion facilitates continuous mass and energy exchange between haloes groups on different mass scales m_h, i.e. an inverse mass and energy cascade (Fig. 1).

The highly localized and over-dense haloes are a major manifestation of nonlinear gravitational collapse [45, 46] and the building blocks of dark matter flow, a counterpart to "eddies" in turbulence. The halo-mediated inverse mass cascade is not present in turbulence, but exists as a local, two-way, and asymmetric process in dark matter flow [1]. The net mass transfer proceeds in a "bottom-up" fashion from small to large mass scales (inverse cascade) to allow for hierarchical structure formation. Haloes pass their mass onto larger and larger haloes, until halo mass growth becomes dominant over mass propagation. From this description, mass cascade can be described by a similar poem with "eddies" (or "whirls") simply replaced by "haloes":

"Little haloes have big haloes, That feed on their mass;
And big haloes have greater haloes, And so on to growth."

Energy cascade across halo groups is facilitated by the mass cascade and is also a fundamental feature. Even on the halo scale, since haloes are non-equilibrium objects, the energy cascade should also play a role in the abundance and internal structure of haloes [47]. In this paper, we focus on the energy cascade, its evidence from galaxy rotation curves, and its critical roles for halo internal structure and dark matter properties.

2 THE CONSTANT RATE OF ENERGY CASCADE

Particle-based N-body simulations are widely used to study the nonlinear gravitational collapse of dark matter [48]. The simulation data for this work was generated from N-body simulations by the Virgo consortium [49, 50]. One way to determine the constant rate of the energy cascade ε_u is from a cosmic energy equation for the energy evolution of dark matter flow in an expanding background [51, 52, 53],

$$\frac{\partial E_y}{\partial t} + H\left(2K_p + P_y\right) = 0, \tag{1}$$

which is a manifestation of energy conservation. Here K_p is the specific (peculiar) kinetic energy, P_y is the specific potential energy in physical coordinate, $E_y = K_p + P_y$ is the total energy, $H = \dot{a}/a$ is the Hubble parameter ($Ht = 2/3$ for the matter-dominant universe), and a is the scale factor. In a statistically steady state, Eq. (1) admits a linear solution of $K_p = -\varepsilon_u t$ and $P_y = 1.4\varepsilon_u t$ (see Fig. 2) such that ε_u can be found as

$$\varepsilon_u = -\frac{K_p}{t} = -\frac{3}{2}\frac{u^2}{t} = -\frac{3}{2}\frac{u_0^2}{t_0} = -\frac{9}{4}H_0 u_0^2 \approx -4.6\times 10^{-7}\frac{m^2}{s^3}, \tag{2}$$

where $u_0 \equiv u(t = t_0) \approx 354.6 km/s$ is the one-dimensional velocity dispersion of all dark matter particles and t_0 is the present age of the simulated universe, $t_0 = 2/(3H_0) = 13.04$ billion years for this run ($h = 0.5$); with $t_0 = 13.8$ billion years Eq. (2) would give $-4.33\times 10^{-7} m^2/s^3$. The constant ε_u represents the rate of the energy cascade across different scales. The negative value $\varepsilon_u < 0$ reflects the direction (inverse) from small to large mass scales. In the simulation the linear solution is approached but not yet reached at $z = 0$: for $a \geqslant 0.5$, $K_p \propto a^{1.23} \propto t^{0.82}$ and $-P_y \propto a^{1.26}$, and $-P_y/K_p = 1.47$ at $z = 0$, close to the value 1.45 that Eq. (1) gives for $K_p \propto t^{0.82}$ (7/5 for $K_p \propto t$). The value of ε_u is therefore a characteristic rate, $K_p(t_0)/t_0$.

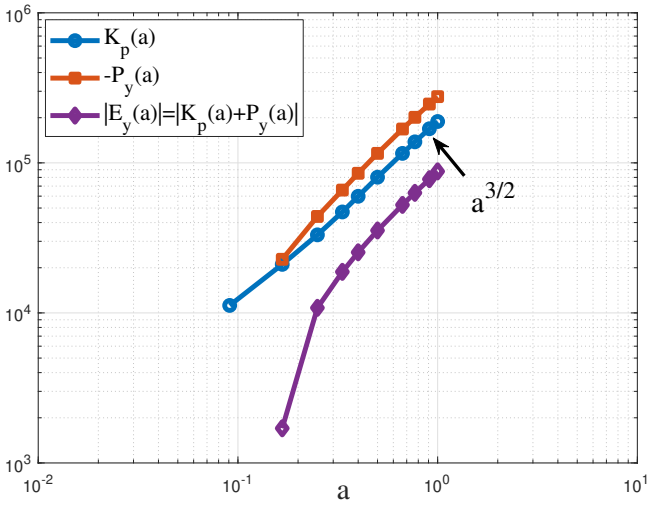


Figure 2. The time variation of kinetic and potential energies from N-body simulation. Both grow nearly as power laws of the scale factor a: $K_p \propto a^{1.23}$ and $-P_y \propto a^{1.26}$ for $a \geqslant 0.5$, slightly slower than the slope $a^{3/2} \propto t$ of the linear solution $K_p = -\varepsilon_u t$ (arrow). The proportional constant ε_u is estimated in Eq. (2).

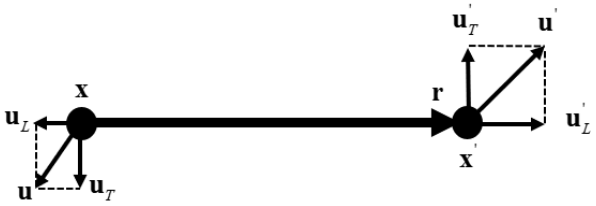


Figure 3. Sketch of longitudinal and transverse velocities, where u_T and u'_T are transverse velocities at two locations x and x′. u_L and u'_L are two longitudinal velocities.

3 THE 2/3 AND -4/3 LAWS FOR ENERGY AND DENSITY

To develop statistical theory of dark matter flow on all scales, different statistical measures can be introduced, including correlation, structure, dispersion functions, and power spectrum for density, velocity and potential fields [43]. Among different measures, structure functions are of particular interest that describe how energy is distributed and transferred across different length scales. For a pair of particles at two different locations x and x′ with velocity u and u′, the second order longitudinal structure function S_2^{lp} (pairwise velocity dispersion in cosmology terms) is defined as

$$S_2^{lp}(r,t) = \left\langle (\Delta u_L)^2 \right\rangle = \left\langle \left(u'_L - u_L\right)^2 \right\rangle, \tag{3}$$

where $u_L = \mathrm{u}\cdot\hat{\mathrm{r}}$ and $u'_L = \mathrm{u}'\cdot\hat{\mathrm{r}}$ are two longitudinal velocities. The distance $r \equiv |\mathrm{r}| = |\mathrm{x}' - \mathrm{x}|$ and the unit vector $\hat{\mathrm{r}} = \mathrm{r}/r$ (see Fig. 3). For a given scale r, all particle pairs with the same separation r can be identified in the N-body simulation. The particle position and velocity data were recorded to compute the structure function in Eq. (3) by averaging over all pairs with the same r (i.e. a pairwise average).

In incompressible flow, the structure function has a small-scale limit $\lim_{r\to 0} S_2^{lp} = 0$ because of $u_L \approx u'_L$ due to the viscous force. However, in dark matter flow, $\lim_{r\to 0} S_2^{lp} = 2u^2 \neq 0$ due to the collisionless nature [54, 55], where u^2 is velocity dispersion in Eq. (2). The pair

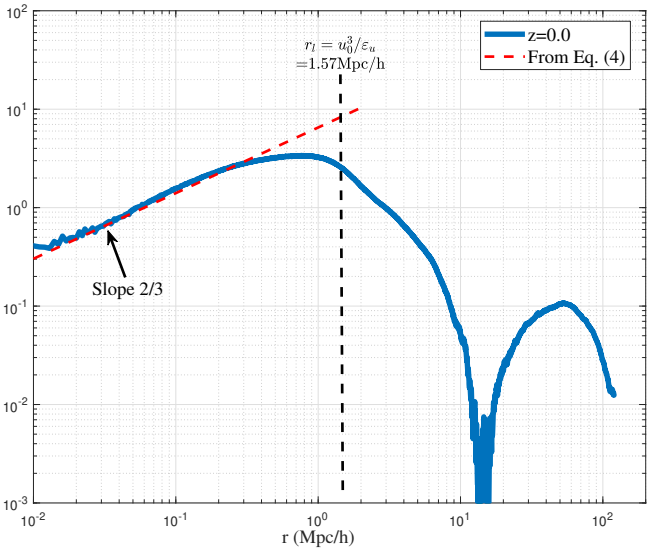


Figure 4. The variation of reduced second order structure function $S_{2r}^{lp}(r)$ with comoving scale r at z=0. Structure function is normalized by the velocity dispersion u^2. A two-thirds law, i.e. $\propto (-\varepsilon_u)^{2/3} r^{2/3}$ can be clearly identified on small scale below a length scale $r_l = -u_0^3/\varepsilon_u$, where inverse energy cascade is established with a constant energy flux $\varepsilon_u < 0$. The model from Eq. (4) is also presented for comparison.

of particles with a sufficiently small r is more likely to be from the same halo, while different pairs can be from different haloes. The kinetic energy of the particle pairs on scale r includes contributions from both the relative motion of two particles and the motion of the haloes in which the particle pair resides. The kinetic energy from the motion of the haloes ($2u^2$) is relatively the same for different pairs. The kinetic energy involved in the energy cascade should be the part due to the relative motion. Since the original structure function (pairwise dispersion) $S_2^{lp}(r)$ includes the total kinetic energy on scale r, a reduced structure function $S_{2r}^{lp}(r) = S_2^{lp} - 2u^2$ can be introduced to take out the common part and include only the part from relative motion with the right limit $\lim_{r\to 0} S_{2r}^{lp} = 0$. This description indicates that $S_{2r}^{lp}(r)$ should be determined by and only by ε_u (m^2/s^3), gravitational constant G ($m^3/kg\cdot s^2$), and scale r. By a simple dimensional analysis, this reduced structure function must follow a two-thirds law, i.e. $S_{2r}^{lp}(r) \propto (-\varepsilon_u)^{2/3} r^{2/3}$.

Figure 4 plots the variation of S_{2r}^{lp} with scale r at z=0 from N-body simulations. The range $S_{2r}^{lp} \propto (-\varepsilon_u r)^{2/3}$ can be clearly identified below a critical length scale $r_l = -u_0^3/\varepsilon_u$. This range is formed by the formation of haloes and the inverse energy cascade. On small scales, S_{2r}^{lp} (kinetic energy) should finally read

$$S_{2r}^{lp}(r) = a^{3/2}\beta_2^*(-\varepsilon_u)^{2/3} r^{2/3}, \tag{4}$$

where the proportional constant $\beta_2^* \approx 9.5$ can be found from Fig. 4 (least-squares fit of the $z = 0$ data over 0.01–0.1 Mpc/h with r in physical units; the red line of Fig. 4 is drawn with 8.8). The data follow Eq. (4) within 20 per cent up to $r \approx 0.5$ Mpc/h; S_{2r}^{lp} peaks at 0.76 Mpc/h, a factor 2 below $r_l = 1.57$ Mpc/h. Since S_{2r}^{lp} represents the kinetic energy of relative motion on scale r, a different form of the two-thirds law (Eq. (4)) can be obtained. By introducing a typical velocity v_r on a given scale r,

$$v_r^2 = S_{2r}^{lp}(r) / \left(2^{2/3}\beta_2^* a^{3/2}\right), \tag{5}$$

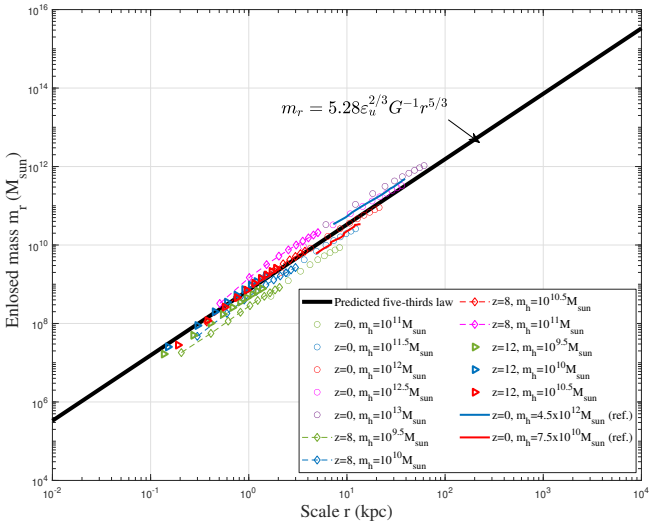


Figure 5. The variation of enclosed mass m_r with scale r at different redshift from Illustris simulation. For all haloes with a given mass m_h, the average scale radius r_s is calculated. The average enclosed mass m_r is calculated for different scales r from $r = 0.1r_s$ to $r = r_s$ with an increase of $0.1r_s$. Solid blue and red lines are results from ref. [57]. Both results follow the predicted five-thirds law in Eq. (7) on average; the slope inside a halo (1.49–1.96) and the coefficient α_r (1.9–10.6) vary with halo mass and redshift.

the two-thirds law in Eq. (4) can be written equivalently as

$$-\varepsilon_u = \frac{2v_r^2}{r}v_r = a_r v_r = \frac{2v_r^2}{r/v_r} = \frac{2v_r^3}{r}, \tag{6}$$

where a_r is the scale of acceleration. Equation (6) also describes the cascade of kinetic energy in the inner halo region ($r < r_s$, where r_s is the scale radius). The kinetic energy v_r^2 on the scale r is cascaded to large scales during a turnaround time of $t_r = r/v_r$. Combining Eq. (6) with the virial theorem $Gm_r/r \propto v_r^2$ on scale r, we can easily obtain the typical mass m_r (enclosed within r), density ρ_r, velocity v_r, and time t_r on scale r, all determined by ε_u, G and r:

$$\begin{aligned} &m_r = \alpha_r \varepsilon_u^{2/3} G^{-1} r^{5/3} \quad \text{and} \quad \rho_r = \beta_r \varepsilon_u^{2/3} G^{-1} r^{-4/3}, \\ &v_r \propto (-\varepsilon_u r)^{1/3} \quad \text{and} \quad t_r \propto (-\varepsilon_u)^{-1/3} r^{2/3}, \end{aligned} \tag{7}$$

where α_r and β_r are two numerical constants. The predicted five-thirds law for mass m_r enclosed in scale r can be directly tested by N-body simulations. In this work, the large-scale cosmological simulation Illustris (Illustris-1-Dark) was selected for comparison [56]. Figure 5 presents the variation of the enclosed mass m_r with scale r at different redshifts z for all haloes with a given mass m_h. Results from ref. [57] are also presented for comparison. Both results follow the predicted five-thirds law in Eq. (7) on average. Inside each halo, however, $m_r \propto r^{1.49\text{–}1.96}$ (Illustris) and $r^{1.59}$, $r^{1.76}$ (ref. [57]), and the coefficient α_r ranges from 1.9 to 10.6 (line: 5.28); at $z = 0$ it grows with halo mass as $\alpha_r \propto m_h^{0.36}$. Next, the predicted four-thirds law $\rho_r(r) \propto r^{-4/3}$ for mean density enclosed in scale r can be directly tested by data from galaxy rotation curves (see Fig. 6).

4 HALO DENSITY SLOPE AND MEAN RADIAL FLOW

On small scales, the inner halo region is assumed to be fully virialized for Eq. (7) to be valid. A vanishing radial flow u_r is expected from the stable clustering hypothesis, i.e., no net stream motion in physical coordinate along the radial direction [43, 58]. With $u_r \equiv 0$, there is no mass, momentum, and energy exchange between different spherical shells. However, simulated haloes are non-equilibrium dynamic objects that might not be fully virialized, whose internal structure should be dependent on the radial flow. The relation we will develop in this section might be useful for the core/cusp controversy.

Since halo density models often involve a scale radius r_s (the length scale of a halo), we may introduce a reduced spatial-temporal variable $x = r/r_s(t) = c(t)r/r_h(t)$, where $c \equiv c(t)$ is the halo concentration and $r_h(t)$ is the virial size for halo of mass $m_h(t)$. The derivatives with respect to t and r can be derived using the chain rule,

$$\frac{\partial}{\partial t} = \frac{\partial}{\partial x}\frac{\partial x}{\partial t} = -\frac{x}{t}\frac{\partial \ln r_s}{\partial \ln t}\frac{\partial}{\partial x} \quad \text{and} \quad \frac{\partial}{\partial r} = \frac{\partial}{\partial x}\frac{\partial x}{\partial r} = \frac{1}{r_s}\frac{\partial}{\partial x}. \tag{8}$$

A general function $F(x)$ can be introduced such that the mass $m_r(r,t)$ enclosed in radius r and halo density $\rho_h(r,t)$ can all be expressed in terms of function $F(x)$,

$$\begin{aligned} &m_r(r,t) = m_h(t)\frac{F(x)}{F(c)}, \\ &\rho_h(r,t) = \frac{1}{4\pi r^2}\frac{\partial m_r(r,t)}{\partial r} = \frac{m_h(t)}{4\pi r_s^3}\frac{F'(x)}{x^2 F(c)}. \end{aligned} \tag{9}$$

The time derivative of $\rho_h(r,t)$ can be obtained from Eq. (9),

$$\frac{\partial \rho_h(r,t)}{\partial t} = \frac{1}{4\pi r^2}\frac{\partial^2 m_r(r,t)}{\partial r \partial t}. \tag{10}$$

The mass continuity equation for a spherical halo simply reads,

$$\frac{\partial \rho_h(r,t)}{\partial t} + \frac{1}{r^2}\frac{\partial\left[r^2\rho_h(r,t)u_r(r,t)\right]}{\partial r} = 0, \tag{11}$$

where $u_r(r,t)$ is the mean radial flow velocity. From Eqs. (10) and (11), the enclosed mass m_r is related to the radial flow as

$$\frac{\partial m_r(r,t)}{\partial t} = -4\pi r^2 u_r(r,t)\rho_h(r,t). \tag{12}$$

With m_r and ρ_h from Eq. (9), the radial flow u_r simply reads

$$u_r = -\frac{1}{4\pi r^2}\frac{\partial \ln m_r}{\partial \ln t}\frac{m_r(r,t)}{\rho_h(r,t)t} = -\frac{r_s(t)}{t}\frac{\partial \ln m_r}{\partial \ln t}\frac{F(x)}{F'(x)}. \tag{13}$$

While from Eq. (9) for m_r, we have

$$\frac{\partial \ln m_r}{\partial \ln t} = \frac{\partial \ln m_h}{\partial \ln t} - \frac{xF'(x)}{F(x)}\frac{\partial \ln r_s}{\partial \ln t} - \frac{cF'(c)}{F(c)}\frac{\partial \ln c}{\partial \ln t}. \tag{14}$$

Substituting Eq. (14) into Eq. (13), the normalized radial flow u_h reads

$$u_h(x) = u_r\frac{t}{r_s} = \left[x\frac{\partial \ln r_s}{\partial \ln t} + \left(\frac{\partial \ln F(c)}{\partial \ln t} - \frac{\partial \ln m_h}{\partial \ln t}\right)\frac{F(x)}{F'(x)}\right]. \tag{15}$$

For fast-growing haloes in their early stage with constant concentration $c = 3.5$ and scale radius $r_s(t) \propto m_h(t) \propto t$, using Eq. (15), the cored density profiles (pISO, Einasto, etc. with $F(x) \propto x^3$ in Eqs. (19) and (20)) lead to $u_h = 2x/3$ for small x, while the cuspy density profile (NFW with $F(x) \propto x^2$ in Eq. (20)) leads to $u_h = x/2$ for small x. Taking the derivative in Eq. (15) and combining with Eq. (9), the density slope γ can be obtained exactly as

$$\gamma = \frac{\partial \ln \rho_h}{\partial \ln x} = \frac{\frac{\partial u_h}{\partial x} + \frac{\partial \ln m_r(r_s,t)}{\partial \ln t} - \frac{\partial \ln r_s}{\partial \ln t}}{\frac{\partial \ln r_s}{\partial \ln t} - \frac{u_h}{x}} - 2. \tag{16}$$

Clearly, the spatial variation of the slope γ comes from the radial flow u_h, while the time variation of γ comes from $r_s(t)$ and $m_r(r_s,t)$ due to mass accretion. For fully virialized haloes or the virialized

inner core, we should have $u_h \equiv 0$ such that the asymptotic slope γ reads

$$\gamma = \frac{\partial \ln \rho_h}{\partial \ln x} = \frac{\partial \ln m_r(r_s,t)}{\partial \ln r_s} - 3. \quad (17)$$

On the halo scale, the energy cascade with a constant rate ε_u is valid for all scales $r \leqslant r_s$. Taking the enclosed mass $m_r(r_s)$ as the mass scale in Eq. (7), we found $m_r(r_s) \propto r_s^{5/3}$. For fully virialized haloes with $u_h \equiv 0$, slope $\gamma = -4/3$ or cuspy density $\rho_h(r) \propto r^{-4/3}$ can be obtained from Eq. (17). Therefore, fully virialized haloes should have universal cuspy density profiles due to the energy cascade. In other words, simulated haloes might have different slopes γ due to nonzero radial flow and different mass accretion rates (Eq. (16)).

The baryonic feedback provides potential mechanisms to enhance the gradient of u_h (deformation rate) in Eq. (16)) and flatten the inner density. To better illustrate, we can approximate Eq. (16) in the core region as:

$$\gamma \approx \frac{\frac{\partial \ln m_r(r_s)}{\partial \ln r_s}}{1-\mu} - 3, \quad \mu = \frac{\partial u_h}{\partial x} \Big/ \frac{\partial lnr_s}{\partial lnt} \approx \frac{u_h}{x} \Big/ \frac{\partial lnr_s}{\partial lnt}. \quad (18)$$

Here, the dimensionless number μ represents the competition between radial flow and halo mass accretion, i.e. the deformation rate normalized by the rate of change in the halo core size. The exponent $m = \partial lnr_s/\partial lnt$ decreases with time and increases with the halo size to $m \approx 1$ for large haloes. From the scaling laws for m_r in Eq.(7), $m_r(r_s) \propto r_s^{5/3}$. From Eq. (18), strong supernova-driven outflows in interstellar medium may lead to a nonzero radial outflow u_h of dark matter or a greater μ such that core density becomes flatter from Eq. (18). This effect should be greater in smaller haloes due to the smaller exponent m. Haloes should also be large enough for a sufficient fraction of baryons converted into stars to allow feedback. Density cores are only developed in a certain range of halo mass (10^{10} to $10^{11} M_\odot$), as confirmed by cosmological simulations [59].

Finally, since $u_h(x=0) \equiv 0$, there exists an asymptotic slope γ for simulated haloes for small $x \to 0$ that depends on the local gradient of u_h around $x = 0$ (see Eq. (16)). Therefore, a better density profile can be proposed for simulated haloes (see [47] for details)

$$\rho_h(r) = \rho_0 x^\gamma \exp\left(-\frac{2}{\alpha}x^\alpha\right), \quad F(x) = \Gamma(\frac{3+\gamma}{\alpha}) - \Gamma(\frac{3+\gamma}{\alpha}, \frac{2}{\alpha}x^\alpha). \quad (19)$$

Here ρ_0 and α are the density and shape parameters and slope $\gamma < 0$. Different simulated haloes might have a different slope γ due to the different behavior of the radial flow u_h at small x and halo mass accretion.

5 TESTING -4/3 LAW AGAINST ROTATION CURVES

Next, the four-thirds law predicted in Eq. (7) ($\rho_r(r_s) \propto r_s^{-4/3}$) is tested against galaxy rotation curves that contain important information for dark matter haloes. In practice, rotational curves can first be decomposed into contributions from different mass components. The model parameters for the halo density (scale radius r_s and density scale ρ_0) are obtained by fitting to the decomposed rotation curve. In this work, we use three sources of galaxy rotation curves,

(i) SPARC (Spitzer Photometry & Accurate Rotation Curves) : 153 of its 175 late-type galaxies (those with a flat rotation velocity in the baryonic Tully–Fisher sample), with the NFW and pISO fits of ref. [60] [60, 61];

(ii) DMS (DiskMass Survey) including 30 spiral galaxies [62];

(iii) SOFUE (compiled by Sofue) with 43 galaxies [63].

The mass modeling of galaxies is always challenging because of the uncertainties in the stellar mass-to-light ratio $\Upsilon_\star$. Both the DiskMass Survey (DMS) and stellar population synthesis models (SPS) suggest an almost constant $\Upsilon_\star$ in the near-infrared band for galaxies of different masses and morphologies [61]. Wang and Chen removed SPARC galaxies that have a bulge component to improve the fitting quality because of a different mass-to-light ratio between the bulge and the disk [64]. In the SPARC halo fits used in this work [60], 31 of the 153 galaxies have a bulge; the mass-to-light ratios of disk and bulge are fit parameters, and the fitted ratios $\Upsilon_{bul}/\Upsilon_{disk}$ range from 0.21 to 2.68 (median 1.1) instead of the fixed SPS ratio 1.4.

For pseudo-isothermal (pISO) and NFW density, we have

$$\rho_{pISO} = \frac{\rho_0}{1+x^2}, \quad F(x) = x - \arctan(x) \approx \frac{x^3}{3}, \quad \rho_{NFW} = \frac{\rho_0}{x(1+x)^2}, \quad F(x) = \ln(1+x) - \frac{x}{(1+x)} \approx \frac{x^2}{2}, \quad (20)$$

where ρ_0 is a density parameter. From Eq. (9), halo density at r_s is

$$\rho_h(r_s) = \rho_h(x=1) = \bar{\rho}_h \frac{c^3 F'(1)}{3F(c)} = \Delta_c \rho_{crit} \frac{c^3 F'(1)}{3F(c)}, \quad (21)$$

where $\bar{\rho}_h$ is the mean halo density. In this work, $\Delta_c = 200$ and critical density $\rho_{crit} = 3H_0^2/8\pi G = 10^{-26} kg/m^3$ ($H_0 = 73$ km/s/Mpc, the value of the SPARC fits, whose M_{200}, c and r_s imply $1.01 \times 10^{-26} kg/m^3$). For an NFW fit Eq. (22) makes ρ_s a function of c alone, so the NFW points of Fig. 6 trace the concentration–mass relation of the fits, which for SPARC include ΛCDM priors [60]. Using Eqs. (20) and (21), the concentration c can be obtained from the parameter of the fitted model ρ_0. The mean density within r_s (density scale ρ_s) now reads

$$\rho_s(r_s) = \frac{m_r(r_s)}{4\pi r_s^3/3} = \frac{F(1)c^3}{F(c)}\bar{\rho}_h = \frac{F(1)c^3}{F(c)}\Delta_c \rho_{crit}, \quad (22)$$

Figure 6 presents the variation of typical density ρ_s with scale r_s obtained from three different sources of galaxy rotation curves. There is a strong correlation between core density and scale radius with a Pearson correlation coefficient of -0.91. The four-thirds law (Eq. (7)) is also plotted for comparison with the coefficients $\beta_r = 1.26$ and $\alpha_r = 5.28$ obtained from these data as the arithmetic mean of $\rho_s r_s^{4/3} G/|\varepsilon_u|^{2/3}$; the geometric mean (and the median) is $\beta_r = 0.86$, i.e. $\alpha_r = 3.6$, so that the plotted line lies 0.16 dex above the centre of the data in this logarithmic plot. The black dashed line shows the least-squares fit to data from all 409 fits of 226 galaxies (NFW and pISO fits of 153 SPARC and 30 DMS galaxies, NFW fits of 43 Sofue galaxies) with the best slope of −1.37 (95 per cent interval −1.43 to −1.31; a bootstrap over galaxies gives −1.44 to −1.30), which is very close to -4/3. The R-square value (the percentage of data variation that can be explained by the model) of the fit is 0.82 with a root mean square scatter of all data of 0.38 dex. The pooled slope depends on the mix of samples and profiles: the individual samples give −1.07 (SPARC NFW), −1.34 (SPARC pISO), −1.69 (DMS NFW), −1.77 (DMS pISO) and −1.56 (Sofue NFW); NFW fits alone give −1.28 and pISO fits alone −1.44. The Einasto fits of the same SPARC galaxies [60] give −1.06, and −1.31 when they replace the SPARC NFW fits in the pool; the bisector of the two least-squares regressions gives −1.51. Previous studies on the scaling between the halo density and core radius have different slopes for different model fits. A constant surface density ($\rho_s r_s$=constant

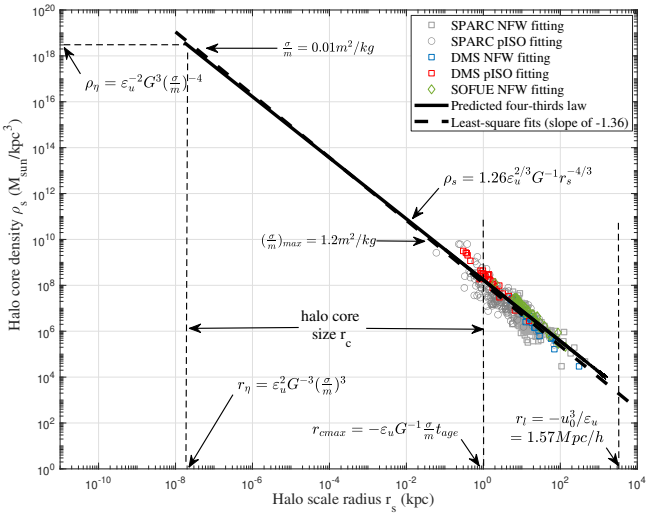


Figure 6. The predicted -4/3 law tested against actual data from galaxy rotation curves. Good agreement confirms the existence of inverse energy cascade with a constant rate ε_u. The self-interacting dark matter model should modify the lowest size r_η and maximum density ρ_η determined by the cross-section σ/m, below which no coherent structure can exist. The largest possible core size r_{cmax} due to self-interaction is determined by the age of haloes t_{age}. The largest scale r_l is determined by the velocity dispersion u_0 and ε_u.

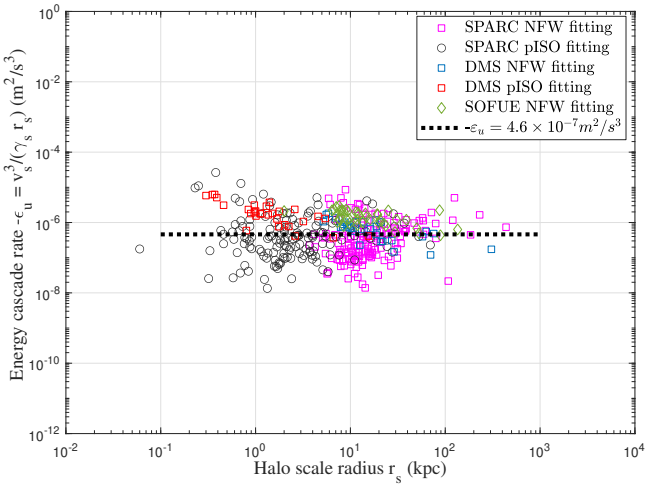


Figure 7. The constant rate of inverse energy cascade ε_u from galaxy rotation curve data. The dispersion in data might come from the spatial intermittence of energy cascade. Smaller haloes have smaller ε_u in the SPARC NFW fits ($\varepsilon_u \propto r_s^{0.40}$), but not in the other samples ($r_s^{0.00}$ for SPARC pISO, $r_s^{-0.34}$ to $r_s^{-0.66}$ for Sofue and DMS); the pooled data give $r_s^{-0.05}$.

or $\rho_s \propto r_s^{-1.04}$) was first observed from 55 rotation curves of spiral galaxies [65] for isothermal fits (ISO). However, the slope becomes -1.20 for pseudo-isothermal (pISO) fits in the same study. Spano et al. also found a slope of -0.93 for ISO fits and -1.0 for NFW fits based on data from 36 spiral galaxies with central density across three orders [66]. Some studies also suggest a non-constant surface density ($\rho_s r_s \neq$const). Examples are the slope of -2 [67] based on 40 spiral galaxies with core density across three orders. A slope of -1.46 was also found in the galaxy cluster [68], which is close to this work.

In this work, we proposed a theory for the existence of such scaling laws. Dark matter haloes obtained from rotation curves follow the predicted four-thirds law across about 4 orders of magnitude in size ($r_s = 0.06$–440 kpc) and 5 in density. Equivalently, the inner cuspy density $\rho_h \propto r^{-4/3}$ for virialized haloes with a vanishing radial flow is also confirmed by this plot or Eq. (17). This plot also confirms the existence of a constant rate of cascade ε_u below the largest halo scale r_l. Other relevant quantities (density, pressure, energy, etc.) on scale r_l are similarly determined by constants ε_u, G, and u_0 [69].

Finally, we can choose the circular velocity at r_s as the typical velocity $v_s = \sqrt{G m_r(r_s)/r_s}$ such that $-\varepsilon_u = v_s^3/(\gamma_s r_s)$ (from Eq. (7)). Figure 7 presents the rate of energy cascade ε_u obtained from three sources of galaxy rotation curves with $\gamma_s = 6.83$, the value for which the geometric mean of the data is $-\varepsilon_u = 4.6 \times 10^{-7} m^2/s^3$. Since $v_s^3/r_s = (4\pi G \rho_s r_s^{4/3}/3)^{3/2}$, Eq. (7) gives $\gamma_s = \alpha_r^{3/2}$: $\gamma_s = 6.83$ corresponds to $\alpha_r = 3.6$ ($\beta_r = 0.86$, the geometric mean of Fig. 6), while $\alpha_r = 5.28$ would give $\gamma_s = 12.1$. Figure 7 is therefore Fig. 6 in other variables: its flat mean expresses the slope $-1.37 \approx -4/3$, and its scatter is 0.57 dex (sample means from 3.1×10^{-7} for SPARC NFW to $1.7 \times 10^{-6} m^2/s^3$ for DMS pISO). The dispersion in data might come from the spatial intermittence of energy cascade such that haloes in different local environment may have slightly different ε_u.

6 SCALES FOR SELF-INTERACTING DARK MATTER

To solve the core-cusp problem, one option is the self-interacting dark matter (SIDM) model, where cascade theory can be used to determine relevant scales. The cross section σ/m of self-interaction should introduce additional scales r_η, ρ_η, and m_η (see Fig. 6), beyond which no structure can be formed due to self-interaction. These scales can be obtained by requiring at least one scatter per particle during the typical time t_r, that is, $\rho_r(\sigma/m)v_r t_r = 1$ in Eq. (7). Combine this with the virial theorem and the constant energy cascade in Eq. (6), the length scale r_η, the density scale ρ_η, and the mass scale m_η are all determined by ε_u, G, and σ/m (see Fig. 6 and Eq. (23)),

$$r_\eta = \varepsilon_u^2 G^{-3} (\sigma/m)^3, \quad m_\eta = \varepsilon_u^4 G^{-6} (\sigma/m)^5, \\ \rho_\eta = \varepsilon_u^{-2} G^3 (\sigma/m)^{-4} \tag{23}$$

From Eq. (23), the upper limit of the cross section can be estimated as

$$\frac{\sigma}{m} \leqslant \left(\frac{r_\eta}{1kpc}\right)^{\frac{1}{3}} \frac{G(1kpc)^{1/3}}{\varepsilon_u^{2/3}} = 35.1 \frac{cm^2}{g} \left(\frac{r_\eta}{1kpc}\right)^{\frac{1}{3}}. \tag{24}$$

From galaxy rotation curves with a maximum core density around $\rho_\eta \approx 10^{10} M_{sun}/(kpc)^3$ and $r_\eta \approx 0.04$kpc (Fig. 6), we can safely estimate the upper limit of $\sigma/m \leqslant 12 cm^2/g$ from Eq. (24) (this point lies on the law with $\beta_r = 1$ at the top of the observed density range; the smallest fitted r_s is 0.06 kpc, and the densest fitted halo, IC 4202 with $r_s = 0.38$ kpc and $\rho_s = 6.3 \times 10^9 M_{sun}/kpc^3$ (pISO), gives $\sigma/m \leqslant 1/(\rho_s r_s) = 2.0 cm^2/g$ when the same one-scatter condition is applied to it). High-resolution rotation curves for dwarf galaxies should provide more stringent constraints on the cross section by determining the smallest length scale r_η. For comparison, the first constraint using colliding galaxy clusters found that $\sigma/m < 5 cm^2/g$ [70]. The analysis performed on the Bullet Cluster leads to an upper limit of $\sigma/m < 4 cm^2/g$ from MACS J0025.4-1222 [71] and $\sigma/m < 7 cm^2/g$ from DLSCL J0916.2+2951 [72]. A more stringent constraint was also obtained from Bullet Clusters 1E 0657-56 with $\sigma/m < 2 cm^2/g$ [73].

In addition, due to self-interaction, dark matter halo should have an isothermal core with a maximum core size r_{cmax} by requiring

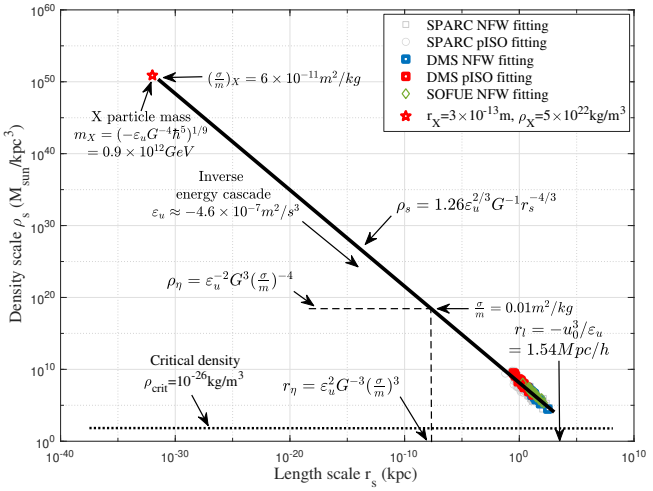


Figure 8. For fully collisionless dark matter, we may extend the predicted -4/3 law in Fig. 6 to the smallest scale where quantum effect can be important (red star). On this scale, the properties of dark matter particles are determined by three constants: ε_u, G, and Planck constant $\hbar$ [54]. The label $r_l = 1.54\,\mathrm{Mpc}/h$ should read $1.57\,\mathrm{Mpc}/h$ (Figs. 4 and 6); the label $m_X = 0.9\times10^{12}\mathrm{GeV}$ is $\mu_X/|\varepsilon_u|$ of Eq. (27), which equals $(-\varepsilon_u G^{-4}\hbar^5)^{1/9}$ times $233^{1/9} = 1.83$.

at least one scatter during the age of haloes ($t_{age} \approx 1/H_0$), that is, $\rho_r(\sigma/m)v_r t_{age} = 1$ such that (using Eq. (7))

$$\frac{r_{cmax}}{\sigma/m} = \varepsilon_u G^{-1} t_{age} = 10kpc\frac{g}{cm^2}. \tag{25}$$

For the cross section $\sigma/m = 0.01m^2/kg = 0.1cm^2/g$ used in SIDM cosmological simulations [33], the halo core size is between r_η and the maximum core size $r_{cmax} \approx 1kpc$. In other words, the maximum core size r_{cmax} formed from self-interaction (not any other mechanism) can be used to identify the cross section of postulated self-interaction.

In hydrodynamic turbulence, the smallest length scale $\eta = (\nu^3/\varepsilon)^{1/4}$ [74] is determined by the viscosity of the fluid ν and the rate of the energy cascade ε. The kinetic energy is injected at large scale and cascaded down to small scales. Below the scale η, the structures (eddies) are destroyed by viscous force, and the kinetic energy is dissipated into heat to increase the entropy of the system (Fig. 1). Here ε can be a variable manually controlled or adjusted by the rate of energy injection on large scales. For faster mixing, thinking of stirring the coffee-milk fluid harder (the larger ε), the scale η would be smaller and mixing would be faster due to the higher velocity on that scale. However, for dark matter flow in our universe, inverse (NOT direct) energy cascade is required for hierarchical structure formation on large scales. The rate of energy cascade ε_u is a constant on small scales $r < r_l$ that cannot be manually controlled, which should be a fundamental constant relevant to dark matter properties. Finally, for self-interacting dark matter, it should be interesting to identify the smallest and greatest halo core size (r_η and r_{cmax}) to constrain the cross-section using Eqs. (24) and (25).

7 SCALES FOR COLLISIONLESS DARK MATTER

Now let us consider the second option: dark matter is fully collisionless with a flat halo core formed by baryonic feedback mechanism. In this scenario, due to the collisionless nature and scale independence of ε_u, the four-thirds law should extend from the galaxy scale to the smallest scale where the quantum effect becomes important (Fig. 8). This extension is more than 30 orders in size, which hopefully allows us to predict the mass, size, and properties of dark matter particles (the X particle) from three basic constants, i.e. ε_u, G, and the Planck constant $\hbar$ [54]. Two examples are the critical mass and length scales:

$$m_X \propto \left(-\frac{\varepsilon_u\hbar^5}{G^4}\right)^{\frac{1}{9}} \approx 10^{12}GeV, \quad r_X \propto \left(-\frac{G\hbar}{\varepsilon_u}\right)^{\frac{1}{3}} \approx 10^{-13}m. \tag{26}$$

If this is true, the constant ε_u might be an intrinsic property of dark matter with a similar origin as Planck constant $\hbar$ for two reasons:

(i) For fully collisionless dark matter, there exists a unique "symmetry" between position and velocity in phase space. At any given location, collisionless particles can have multiple values of velocity (multistream regime). Similarly, particles with the same velocity can be found at different locations. This "symmetry" in phase space is not possible for any nonrelativistic baryonic matter.
(ii) Due to the long-range gravitational interaction, there exist fluctuations (uncertainty) not only in position ($\mathbf{x}$) and velocity ($\mathbf{v} = \dot{\mathbf{x}}$), but also in acceleration ($\mathbf{a} = \dot{\mathbf{v}}$) [69].

With $\psi(x)$, $\varphi(p)$, and $\mu(a)$ as wave functions for position, momentum, and acceleration, we can write

$$\begin{aligned}\psi(x) &= \frac{1}{\sqrt{2\pi\hbar}}\int_{-\infty}^{\infty}\varphi(p)\cdot e^{ip\cdot x/\hbar}\,dp\,,\\ \varphi(p) &= \frac{1}{\sqrt{2\pi\mu_X}}\int_{-\infty}^{\infty}\mu(a)\cdot e^{ia\cdot p/\mu_X}\,da\,,\end{aligned} \tag{27}$$

where constant $\mu_X = -m_X\varepsilon_u = 7.44\times10^{-22}kg\cdot m^2/s^3$ ($m_X = 1.62\times10^{-15}kg = 0.91\times10^{12}$GeV; Eq. (26) with prefactor 1 gives 4.9×10^{11}GeV). An energy scale $\sqrt{\hbar\mu_X} \approx 10^{-9}eV$ ($1.7\times10^{-9}eV$) can be obtained for the possible dark radiation due to dark matter annihilation or decay. Here we have two pairs of conjugate variables: i) position x and momentum p, and ii) momentum p and acceleration a. By following the standard wave mechanics (will not repeat here), two uncertainty principles can be established for fluctuations of position, momentum, and acceleration for collisionless dark matter,

$$\sigma_x\sigma_p \geqslant \hbar/2 \quad \text{and} \quad \sigma_p\sigma_a \geqslant \mu_X/2. \tag{28}$$

More experimental data might be required to test this postulation.

8 CONCLUSIONS

Small scale challenges suggest some missing pieces in our current understanding of dark matter. A cascade theory for dark matter flow provides extra insights. The energy cascade with a constant rate ε_u across different scales is a fundamental feature of dark matter flow. The simulation of N-body suggests a two-thirds law, i.e., the kinetic energy $v_r^2 \propto (\varepsilon_u r)^{2/3}$ on scale r. This is equivalent to a four-thirds law for density on the same scale, i.e. $\rho_r \propto \varepsilon_u^{2/3}G^{-1}r^{-4/3}$, which can be directly confirmed by data from N-body simulations and galaxy rotation curves. By identifying key constants on relevant scales, limiting scales for collisionless (determined by ε_u, $G, \hbar$) or self-interacting dark matter (by ε_u, G, σ/m) might be obtained. On the halo scale, based on the continuity equation, the halo density is shown to be closely dependent on the radial flow and mass accretion. The asymptotic density slope $\gamma = -4/3$ can be obtained for fully virialized haloes with vanishing radial flow. Simulated haloes

can have different limiting slopes due to finite radial flow and different rates of mass accretion. The baryonic feedback might enhance the radial flow and flatten the core density. A modified Einasto density profile is proposed accordingly.

DATA AVAILABILITY

Two datasets for this article, i.e. halo-based and correlation-based statistics of dark matter flow, are available on Zenodo [75, 76], along with the accompanying presentation "A comparative study of dark matter flow & hydrodynamic turbulence and its applications" [41].

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